

Dhritish Mondal

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4th Year Undergraduate

Department of Electronics and Electrical Communication Engineering

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2028	B.Tech+M.Tech (Dual Degree)	Indian Institute of Technology, Kharagpur	8.14/10
2022	ISC (XII)	Calcutta Public School, Kolkata	85%
2020	ICSE (X)	Ram Mohan Mission School, Kolkata	90%

KEY PROJECTS

Hand Gesture Recognition Control Car (DIY project) *Prof. Sanjeev Srivastava* Jan'24–Mar'24

- Developed a **Hand Gesture Recognition Controlled Car** using Python, implementing advanced computer vision techniques to ensure high accuracy under varying environmental conditions.
- Optimized power consumption** to enhance the car's long-term usability, balancing performance and energy efficiency for extended operation.
- Integrated real-time gesture recognition algorithms** Integrated real-time gesture recognition algorithms using OpenCV and machine learning, improving system responsiveness and robustness in diverse lighting and background scenarios.
- Integrated hardware–software** pipeline connecting vision algorithms with car actuation, gaining practical experience in embedded systems, IoT, and human–computer interaction applications, successfully demonstrated the project in multiple test environments, highlighting its potential applications in assistive technology, robotics, and human–computer interaction.

IPL-Win Predictive App (Self Project)

May'25–June'25

- Developed an IPL-WIN Predict Web App using ML to forecast match outcomes based on historical IPL data.
- Implemented a Random Forest-based model with feature engineering and hyperparameter tuning.
- Used contour detection, convex hull analysis, and background filtering for robust gesture detection.

TECHNICAL SKILLS

Programming: C | C++ | LaTeX | MATLAB | HTML | CSS | Python | Java | Machine Learning

Frameworks: NumPy | Pandas | Scikit-learn | Flask | Matplotlib

Tools: Jupyter | Colab | VS Code | GitHub | MS Word | PowerPoint | Excel | Power BI

Software/Tools: VS Code | SQL | Jupyter Notebook

POSITIONS OF RESPONSIBILITY

Design Team Member, Spring Fest '24

- Collaborated with the design team to create event props, stage elements, and promotional materials.
- Applied Photoshop and video editing to design engaging event assets.
- Assisted in crowd management during the festival to ensure a safe and organized experience for attendees
- Assisted sponsorship team, secured Rs.80,000 funding

Secretary of Sports — AJCB Hall of Residence, IIT Kharagpur '25

- Led hall sports activities, managing 100+ students across Cricket, Football, Chess, and Athletics.
- Represented hall in Inter-Hall Football Championship.
- Secured 2nd place in Football Men and Bronze in Cricket Men in GC '23.

EXTRA-CURRICULAR ACTIVITIES

- Participated in the Inter IIT Football '24.
- Represented hall in Inter-Hall GC Football Tournament '25.
- Represented hall in Inter-Hall GC Cricket Tournament '25.
- Play Flute and Guitar during leisure, contributing to cultural activities.